

Surface Tension Effects on Surface Instabilities of Dielectric Elastomers

Saman Seifi^a, Harold S. Park^b

^a*Department of Mechanical Engineering, Boston University, Boston, MA 02215*

^b*Department of Mechanical Engineering, Boston University, Boston, MA 02215*

Abstract

Dielectric elastomers have recently been proposed for various biologically-relevant applications, in which they may operate in fluidic environments where surface tension effects may have a significant effect on their stability and reliability. Here, we present a theoretical analysis coupled with computational modeling for a generalized electromechanical analysis of surface stability in dielectric elastomers accounting for surface tension effects. For mechanically deformed elastomers, significant increases in critical strain and instability wavelength are observed for small elastocapillary numbers. Because surface tension penalizes the sharp, short-wavelength crease far more strongly than the smooth wrinkle, it suppresses the subcritical creasing instability above a critical elastocapillary number; beyond this threshold wrinkling becomes the first instability, which is the regime in which our linear perturbation analysis is physically realized. When the elastomers are deformed electrostatically, both surface tension and the amount of pre-compression are found to substantially increase the critical electric field while decreasing the instability wavelength.

Keywords: Wrinkling Instability, Elastocapillary, Perturbation Analysis

1. Introduction

Because soft materials like gels and elastomers often operate under states of compression, there have been significant efforts to examine their mechanical stability under such loading (?????). Recently, efforts have focused on examining the stability of surfaces due to the formation of instabilities such as wrinkles and creases under compression (?????). The studies of wrinkling instabilities have typically been analytical in nature, based on linear perturbation analysis, which enables predictions of critical strains and instability wavelengths in these soft structures. These studies have also been insightfully combined with experiments (???), and also numerical (finite element) analyses (?????).

More recently, linear perturbation analyses have been applied to an interesting class of soft, active materials - dielectric elastomers (DEs). These materials have drawn significant interest due to the large deformations they can undergo resulting from applied electrostatic loading (????). Furthermore, DEs often form surface instabilities like creasing, wrinkling and cratering, which have been studied via experiment, theory and computation (????Shivapooja et al., 2013; ?; ?; ?; ?), and also using analytic techniques based on stability analyses (??) and the nonlinear field theory of ?.

While DEs have drawn significant interest in recent years, many promising technological innovations, such as generating electromechanical motion during magnetic resonance imaging (Carpi et al., 2008), operating soft, underwater robots (Rus and Tolley, 2015; Kim et al., 2013), manipulating microfluidic flow (Holmes et al., 2013), focusing a tunable lens (Carpi et al., 2011), the preparation of bioinspired surfaces (Shivapooja et al., 2013), underwater grippers (Laschi et al., 2012) and soft body locomotion (Marchese et al., 2014), involve operation of the DE in fluidic environments, where elastocapillary effects due to surface tension can become prominent (??). While prior works have examined either the effects of surface tension alone on surface instabilities in elastomers (?) and hydrogels (Kang and Huang, 2010), or the coupled effects of

URL: samansei@bu.edu (Saman Seifi), parkhs@bu.edu (Harold S. Park)

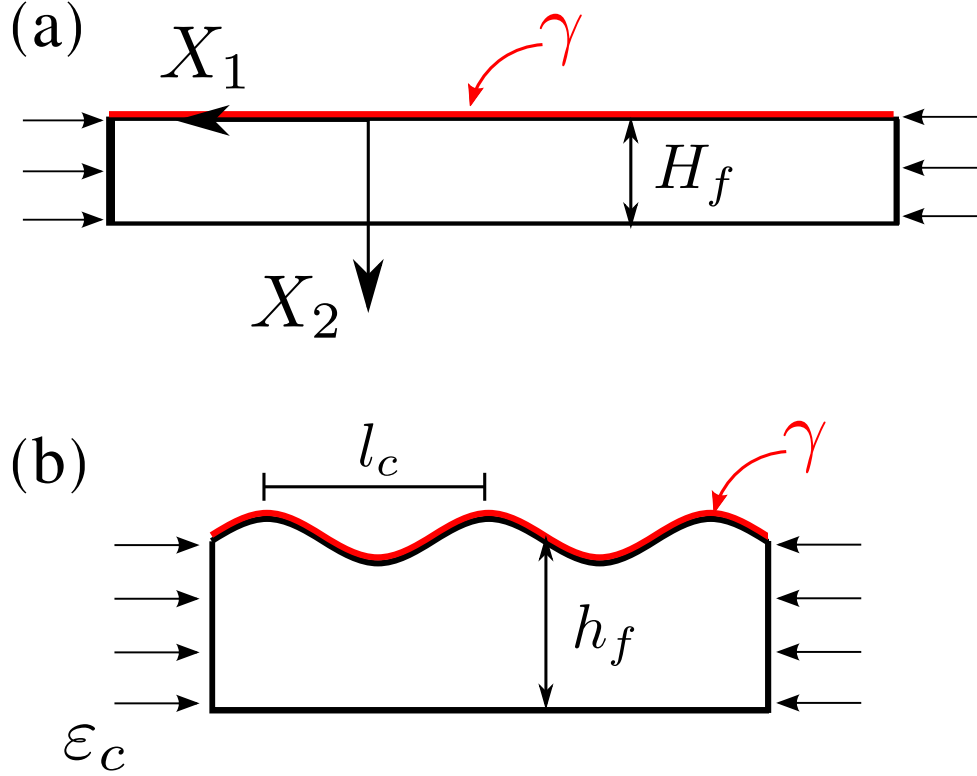


Figure 1: A perfectly flat film subject to compression and surface tension γ . (a) Undeformed configuration; (b) the formation of wrinkles with wavelength l_c .

surface tension and electric fields on surface instabilities in constrained DE films (Seifi and Park, 2016; ?), a general stability analysis of compressed DE films subject to both surface tension and electric fields has not been performed.

The objective of this work is to present a theoretical analysis coupled with computational modeling for surface tension effects on surface instabilities in DEs, with particular emphasis on surface instability wavelengths, and critical strains or electric fields needed to induce the surface instabilities. We consider first the case of surface tension effects on a mechanically compressed DE film, then move on to consider the general case of electrostatic loading on a mechanically compressed DE film.

2. Governing Equations and Incremental Forms

We first present, for a DE film subject to surface tension and under compressive loading, the governing equations and their incremental forms. We assume that the surface of the film is subject to Young-Laplace boundary conditions in order to account for elastocapillary effects on the surface (?). We initially neglect the effects of applied electrostatic loading (i.e. voltages or electric fields), which will be accounted for in the subsequent section. The perturbation analysis we utilize is based on those performed by Kang and Huang (2010); ?; ?

2.1. Dielectric Elastomer Subject to Surface Tension

We first consider a DE film under compression, that is also subject to surface tension effects on its top surface, as illustrated in Figure (1). The deformation of the body is described by the field $\mathbf{x}(\mathbf{X})$. The deformation gradient tensor $\mathbf{F}$ is defined as:

$$\mathbf{F} = \nabla \mathbf{x} \quad (1)$$

The DE is modeled as an incompressible neo-Hookean material with the free energy density function:

$$W(\mathbf{F}) = \frac{\mu}{2} \mathbf{F} : \mathbf{F} - \pi(\det \mathbf{F} - 1) \quad (2)$$

where $\mathbf{A} : \mathbf{B} = \text{tr}(\mathbf{A}^T \mathbf{B})$ denotes the tensor inner product. The first Piola-Kirchhoff (nominal) stress tensor $\mathbf{S}$ can be derived as:

$$\mathbf{S} = \frac{\partial W(\mathbf{F})}{\partial \mathbf{F}} = \mu \mathbf{F} - \pi \mathbf{F}^{-T} \quad (3)$$

where μ is the shear modulus, and $\pi = \pi(\mathbf{X})$ is the Lagrange multiplier to enforce the constraint of incompressibility: $J = \det \mathbf{F} = 1$. The nominal stress satisfies the reference equilibrium equation:

$$\nabla \cdot \mathbf{S} = \mathbf{0} \quad (4)$$

On the surface of the body the film is subjected to the Young-Laplace boundary condition (?) with surface tension γ :

$$\mathbf{S} \mathbf{N} dA = 2\gamma \kappa \mathbf{n} da \quad (5)$$

where $\mathbf{N}$ is the unit normal vector to the surface of the body in the undeformed state, $\mathbf{n}$ is the unit normal vector in the current state, and κ is the mean curvature of the surface.

We now add a perturbation into the state of finite deformation with position $\mathbf{x}_0(\mathbf{X})$, deformation gradient $\mathbf{F}_0(\mathbf{X})$ and nominal stress $\mathbf{S}_0(\mathbf{X})$. If the deformation in the z -direction is constrained, only 2D plane strain deformation is considered. The background deformation gradient of this state is

$$\mathbf{F}_0 = \begin{bmatrix} \lambda & 0 \\ 0 & 1/\lambda \end{bmatrix} \quad (6)$$

Therefore the corresponding base position is $\mathbf{x}_0 = \mathbf{F}_0 \mathbf{X}$. Now by adding a small displacement perturbation $\delta \mathbf{x}$ and pressure perturbation $\delta \pi$, we write the perturbed state as $\mathbf{x} = \mathbf{x}_0(\mathbf{X}) + \delta \mathbf{x}(\mathbf{X})$ and $\pi = \pi_0 + \delta \pi$. Both the base state and the perturbed state satisfy the same governing equations in (1) and (4). The incremental form of the deformation gradient reduces to

$$\delta \mathbf{F} = \nabla(\delta \mathbf{x}) \quad (7)$$

and the equilibrium equation to

$$\nabla \cdot (\delta \mathbf{S}) = \mathbf{0} \quad (8)$$

where the incremental nominal stress tensor can be obtained using directional differentiation at $\mathbf{F}_0$ of the stress in (3):

$$\delta \mathbf{S} = \mu \delta \mathbf{F} - \delta \pi \mathbf{F}_0^{-T} + \pi_0 \mathbf{F}_0^{-T} (\delta \mathbf{F})^T \mathbf{F}_0^{-T} \quad (9)$$

Inserting (9) into (8) we can obtain the following equation:

$$\mu \nabla^2(\delta \mathbf{x}) + \pi_0 \mathbf{F}_0^{-T} \nabla \cdot [(\delta \mathbf{F})^T \mathbf{F}_0^{-T}] - \mathbf{F}_0^{-T} \nabla(\delta \pi) = \mathbf{0} \quad (10)$$

Taking the directional derivative of the incompressibility condition yields:

$$\mathbf{F}_0^{-T} : \delta \mathbf{F} = 0 \implies \frac{1}{\lambda} \frac{\partial(\delta x_1)}{\partial X_1} + \lambda \frac{\partial(\delta x_2)}{\partial X_2} = 0 \quad (11)$$

where the perturbation of the boundary condition reads:

$$\delta \mathbf{S} \mathbf{N} = 2\gamma \delta \kappa \mathbf{F}_0^{-T} \mathbf{N} \quad (12)$$

where the baseline flat state satisfies $\kappa = 0$. At the bottom of the film ($X_2 = H_f$), we have the constrained sliding or fixed boundary conditions:

$$\delta S_{12} = 0 \quad \text{and} \quad \delta x_2 = 0 \quad (13)$$

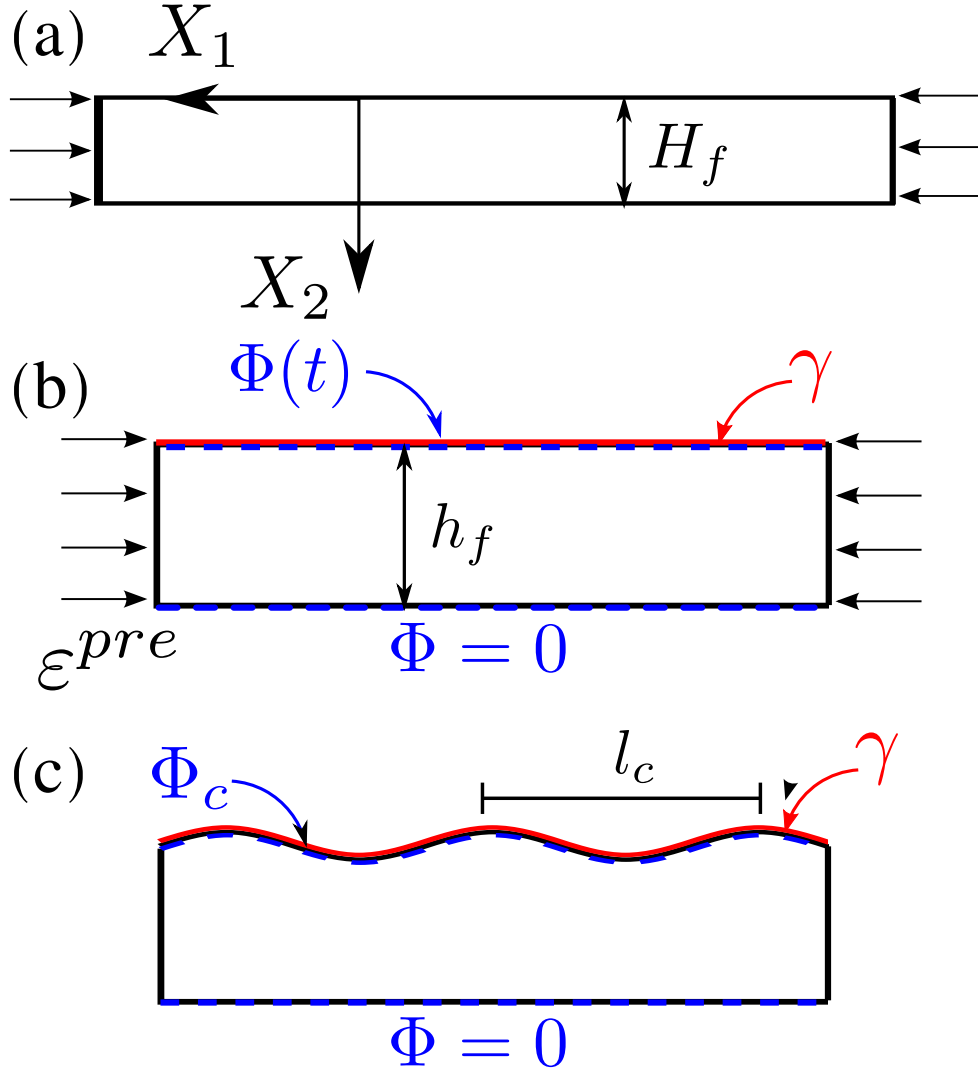


Figure 2: A perfectly flat film subject to compression, electric field and surface tension γ . (a) Undeformed configuration; (b) Compressed to a given strain ε^{pre} , after which voltage $\Phi(t)$ is applied to the top surface while $\Phi = 0$ on the bottom surface; (c) The formation of wrinkles with critical wavelength l_c at critical electric potential Φ_c .

2.2. Dielectric Elastomer with Surface Tension and Electric Field Effects

We now consider the general problem of interest, that of a DE film that is subject to compressive loading, while accounting for both surface tension and electrostatic loading via electric fields. This is illustrated in Figure (2). In this case, the reference configuration is a pre-compressed plane strain film, subject to the background deformation gradient $\mathbf{F}_0$ governed by the mechanical pre-stretch λ^{pre} .

At this state (Figure 2-b), an electric potential Φ is applied on top of the film along with the surface tension γ . When the elastomer is at the flat state, the true electric field in the layer is homogeneous, expressed as:

$$\mathbf{E} = \begin{bmatrix} 0 \\ E_2 \end{bmatrix} \quad (14)$$

where $E_2 = \Phi/h_f$ is the applied electric field and $h_f = H_f/\lambda^{pre}$ is the current height of the pre-compressed film.

The total nominal stress is augmented by the nominal Maxwell electrostatic stress tensor $\mathbf{S}^e$:

$$\mathbf{S}_{total} = \mathbf{S} + \mathbf{S}^e, \quad \mathbf{S}^e = \det(\mathbf{F}) \boldsymbol{\sigma}^e \mathbf{F}^{-T} \quad (15)$$

where $\boldsymbol{\sigma}^e = \epsilon \mathbf{E} \otimes \mathbf{E} - \frac{1}{2} \epsilon |\mathbf{E}|^2 \mathbf{I}$ represents the true spatial Maxwell stress tensor, and ϵ is the material permittivity. Crucially, enforcing the traction-free boundary condition at the flat top face prior to the onset of wrinkling identifies the background hydrostatic pressure π_0 :

$$S_{22}^{total,0} = \left(\frac{\mu}{\lambda^{pre}} - \pi_0 \lambda^{pre} \right) + \frac{1}{2} \lambda^{pre} \epsilon E_2^2 = 0 \implies \pi_0 = \frac{\mu}{(\lambda^{pre})^2} + \frac{1}{2} \epsilon E_2^2 \quad (16)$$

The total stress equilibrium satisfies $\nabla \cdot \mathbf{S}_{total} = \mathbf{0}$. We introduce our field updates using the standard displacement perturbation $\delta \mathbf{x}$ and constraint pressure perturbation $\delta \pi$ mapped directly back to the original reference coordinate frame $\mathbf{X}$ to prevent coordinate loss. The resulting total incremental stress state becomes:

$$\delta \mathbf{S}_{total} = \mu \delta \mathbf{F} - \delta \pi \mathbf{F}_0^{-T} + \pi_0 \mathbf{F}_0^{-T} (\delta \mathbf{F})^T \mathbf{F}_0^{-T} + \delta \mathbf{S}^e \quad (17)$$

The mechanical equilibrium governing equations evaluate to:

$$\left(\mu \delta_{ij} \delta_{KL} + \pi_0 F_{0,iL}^{-T} F_{0,jK}^{-T} \right) \frac{\partial^2 (\delta x_j)}{\partial X_K \partial X_L} - F_{0,iL}^{-T} \frac{\partial (\delta \pi)}{\partial X_K} = 0 \quad (18)$$

where keeping track of the true baseline pre-stress value π_0 is essential to maintain full analytical consistency. On the top boundary face ($X_2 = 0$), the electromechanical traction balances as:

$$\delta \mathbf{S}_{total} \mathbf{N} = 2\gamma \delta \kappa \mathbf{F}_0^{-T} \mathbf{N} \quad (19)$$

where expanding the local Maxwell boundary shift under surface perturbations of height δx_2 provides the stabilizing electrical pressure expression:

$$\delta \boldsymbol{\sigma}^e = \begin{bmatrix} \epsilon E_2^2 \left(\frac{\delta x_2}{h_f} \right) & 0 \\ 0 & -\epsilon E_2^2 \left(\frac{\delta x_2}{h_f} \right) \end{bmatrix} \quad (20)$$

And finally, the boundary conditions at the bottom of the film are fixed against the grounded substrate:

$$\delta x_1(X_1, H_f) = \delta x_2(X_1, H_f) = 0 \quad (21)$$

3. Linear Perturbation Analysis for Wrinkles

3.1. Compression with elastocapillary effect

To determine the onset of wrinkling, we look for non-trivial solutions to the eigenvalue problem via Fourier representation:

$$\begin{aligned}\delta x_1(X_1, X_2) &= f_1(X_2) \sin(KX_1) \\ \delta x_2(X_1, X_2) &= f_2(X_2) \cos(KX_1) \\ \delta \pi(X_1, X_2) &= f_3(X_2) \cos(KX_1)\end{aligned}\tag{22}$$

Substituting these fields into the mechanical equilibrium lines and eliminating the pressure envelope $f_3(X_2)$ yields the classical fourth-order ordinary differential equation for the vertical envelope function f_2 :

$$\frac{d^4 f_2}{dX_2^4} - K^2 (\lambda^{-4} + 1) \frac{d^2 f_2}{dX_2^2} + K^4 \lambda^{-4} f_2 = 0\tag{23}$$

where $K = 2\pi/L$ is the reference wavenumber. The general solution represents an exponential expansion system $\mathbf{AC} = \mathbf{0}$, requiring:

$$\det \mathbf{A} = 0\tag{24}$$

The correct, balanced matrix entries for $\mathbf{A}$ are given as follows:

$$\begin{aligned}A_{11} &= -\frac{\gamma KH_f}{\lambda \mu} - H_f K \left(\lambda^2 + \frac{1}{\lambda^2} \right), & A_{12} &= -\frac{\gamma KH_f}{\lambda \mu} + H_f K \left(\lambda^2 + \frac{1}{\lambda^2} \right) \\ A_{13} &= -\frac{\gamma K^2 H_f}{\mu \lambda} - 2KH_f, & A_{14} &= -\frac{\gamma K^2 H_f}{\mu \lambda} + 2KH_f \\ A_{21} &= A_{22} = \frac{2KH_f}{\lambda^2}, & A_{23} &= A_{24} = \lambda^2 KH_f + \frac{KH_f}{\lambda^2}\end{aligned}\tag{25}$$

$$\begin{aligned}A_{31} &= \frac{2KH_f e^{-KH_f \lambda^{-2}}}{\lambda^2}, & A_{32} &= \frac{2KH_f e^{KH_f \lambda^{-2}}}{\lambda^2} \\ A_{33} &= \left(\lambda^2 KH_f + \frac{KH_f}{\lambda^2} \right) e^{-KH_f}, & A_{34} &= \left(\lambda^2 KH_f + \frac{KH_f}{\lambda^2} \right) e^{KH_f} \\ A_{41} &= e^{-KH_f \lambda^{-2}}, & A_{42} &= e^{KH_f \lambda^{-2}}, & A_{43} &= e^{-KH_f}, & A_{44} &= e^{KH_f}\end{aligned}\tag{26}$$

By solving the eigenvalue problem in (24), we obtain the relation between stretch λ , and compressive strain $\varepsilon = 1 - \lambda$ with the wavelength l (Figure 3).

When surface tension is neglected ($\gamma = 0$), we recover Biot's value of wrinkling at $\varepsilon_w \approx 0.46$ as shown in Figure 3 (?). When surface tension is present ($\gamma > 0$), the strain ε first decreases and then increases as the normalized wavelength l/H_f increases (Figure 3). The key effect of surface tension is that it inhibits bifurcation, as both the critical strain ε_c and corresponding normalized wavelength l_c/H_f increase with increasing $\gamma/(\mu H_f)$ (Figure 4).

In addition, we find that for large elastocapillary numbers, the critical strain approaches a limiting value of $\varepsilon_c \approx 0.85$. The largest change in critical strain occurs for small elastocapillary numbers, i.e. $0 \leq \gamma/(\mu H_f) \leq 2$, where the change in ε_c is more than 34% from $\varepsilon_c = 0.46$ for $\gamma/(\mu H_f) = 0$ to $\varepsilon_c \approx 0.7$ for $\gamma/(\mu H_f) = 2$, while for $\gamma/(\mu H_f) > 2$ the increase in ε_c is less than 18% (Figure 4a).

We verified the theoretical model by performing dynamic, nonlinear finite element (FE) calculations using open source simulation code ? with standard 4-node, bilinear quadrilateral plane strain elements. Simulations were performed on an elastomer film with length $L_f = 40$ and height $H_f = 4$ with a mesh size of $d = 1/16$. As shown in Figure 4, the FE results match the corrected theoretical model very closely.

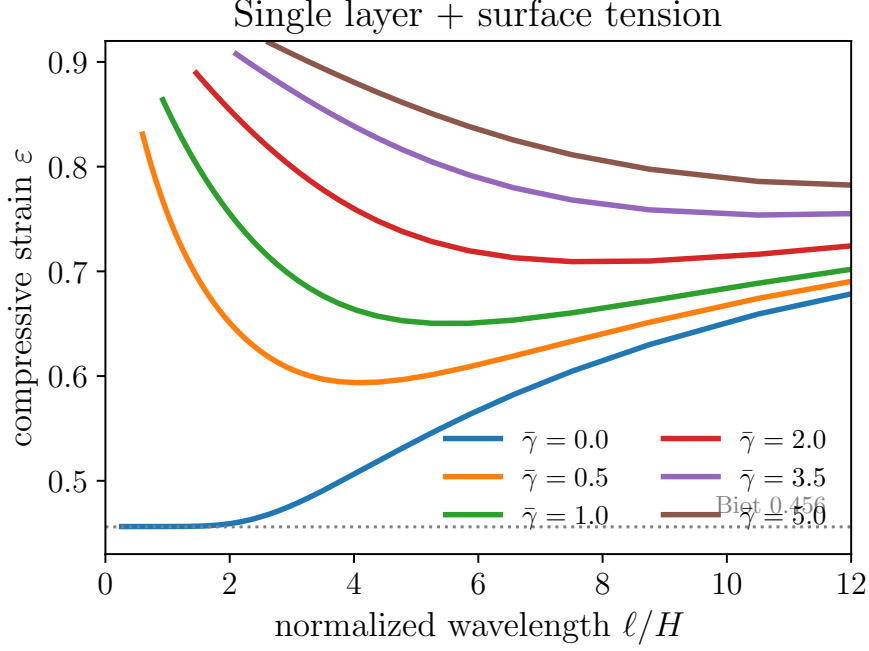


Figure 3: Strain vs. wavelength for a range of elastocapillary numbers

3.2. Crease–wrinkle competition and the validity of the wrinkling analysis

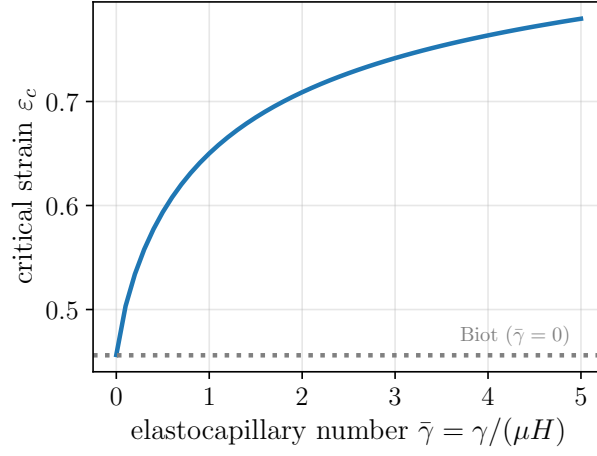
The linear perturbation analysis above predicts the onset of *wrinkling*, a supercritical, smoothly-undulating bifurcation of the flat surface. For a traction-free neo-Hookean half-space this analysis recovers the Biot threshold $\varepsilon_w \approx 0.46$ ($\lambda_w \approx 0.54$). However, a free soft surface under compression is also susceptible to *creasing*: a strongly nonlinear, subcritical, scale-free instability in which the surface folds against itself in a localized self-contact (????). Creasing nucleates at a lower compressive strain, $\varepsilon_{cr} \approx 0.35$ ($\lambda_{cr} \approx 0.65$), and because it is a finite-amplitude instability with no intrinsic wavelength it cannot be captured by the linear (infinitesimal-amplitude) perturbation analysis. Consequently, in the absence of surface tension the crease preempts the wrinkle ($\varepsilon_{cr} < \varepsilon_w$), and the linear wrinkling threshold is an upper bound that is never physically realized.

Surface tension changes this picture qualitatively. A crease is a region of very high surface curvature, so its formation incurs a large surface-energy penalty $\sim \gamma$ per unit fold length; creasing is therefore suppressed, and the crease strain $\varepsilon_{cr}(\gamma/(\mu H_f))$ rises with increasing elastocapillary number. Crucially, the crease—being a sharp, short-wavelength feature—is penalized by surface tension far more strongly than the smooth, long-wavelength wrinkle. As a result ε_{cr} increases faster with $\gamma/(\mu H_f)$ than the wrinkling threshold ε_c obtained in Figure 4(a), and the two cross at a critical elastocapillary number $(\gamma/(\mu H_f))^*$:

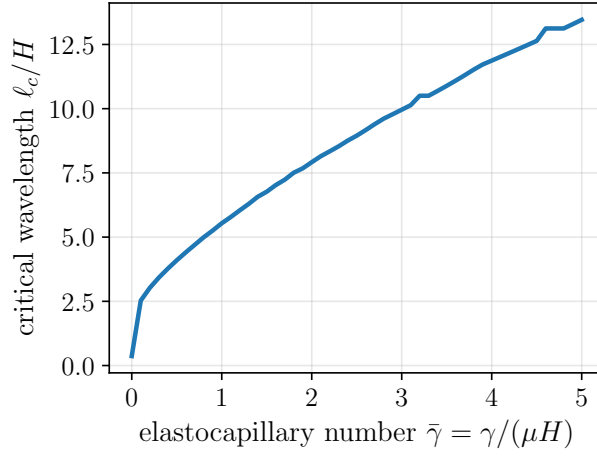
$$\varepsilon_{cr}\left(\frac{\gamma}{\mu H_f}\right) = \varepsilon_c\left(\frac{\gamma}{\mu H_f}\right) \implies \left(\frac{\gamma}{\mu H_f}\right)^*. \quad (27)$$

For $\gamma/(\mu H_f) > (\gamma/(\mu H_f))^*$, the crease is suppressed above the wrinkling threshold, so *wrinkling becomes the first instability encountered as the strain is increased*, and the linear perturbation analysis is the physically relevant prediction. For $\gamma/(\mu H_f) < (\gamma/(\mu H_f))^*$, the subcritical crease still preempts the wrinkle and the linear analysis describes only a latent, unreached bifurcation.

Because creasing is inaccessible to the linear theory, we locate the crossover (27) and the boundaries of the two regimes using the nonlinear finite-element model, in which the Young–Laplace traction is applied directly to the deforming surface and the subcritical crease is allowed to nucleate from a small surface



(a)



(b)

Figure 4: (a): Critical strain ε_c vs. elasto-capillary number $\gamma/(\mu H_f)$; (b): Critical wavelength l_c/H_f vs. elasto-capillary number $\gamma/(\mu H_f)$

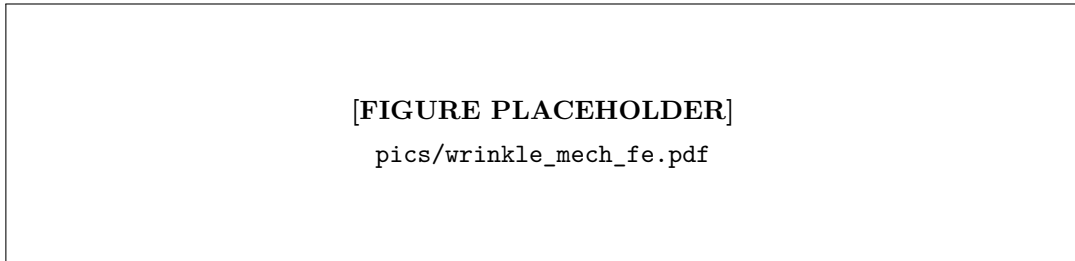


Figure 5: Results for FE simulation of a mechanical compression-induced instability of an 40×4 elastomer film. (a) Undeformed film (b) Onset of wrinkling instability on the top surface at the critical strain $\varepsilon_c = 0.547$ and elastocapillary number $\gamma/(\mu H_f) = 0.2$. $_D_VEC$ is the displacement magnitude.

[FIGURE PLACEHOLDER]

pics/crease_wrinkle_map.pdf

Figure 6: Crease–wrinkle regime map in the strain–elastocapillary plane. Below the crossover $(\gamma/(\mu H_f))^*$ the subcritical crease (strain ε_{cr} , dashed) is the first instability; above it the smooth wrinkle (strain ε_c , solid) is reached first. Markers are nonlinear FE results; the shaded region indicates the surface-tension–dominated wrinkling regime analyzed in this work.

perturbation. The resulting regime map is shown in Figure 6. All of the wrinkling results reported in this work correspond to the surface-tension–dominated regime $\gamma/(\mu H_f) > (\gamma/(\mu H_f))^*$, where the wrinkle is the first instability and the perturbation analysis applies.

3.3. Compressed film subject to surface tension and electric field

To account for electromechanical coupling on the pre-strained layer without losing the background mechanical configuration pressure forces, substituting equations (22) into the coordinate-tracked incremental equilibrium line (18) yields the corrected electromechanical governing differential equation:

$$\frac{d^4 f_2}{dX_2^4} - K^2 \left[\lambda_{pre}^{-4} + 1 + \frac{\epsilon E_2^2}{\mu} (\lambda_{pre}^{-2} - \lambda_{pre}^2) \right] \frac{d^2 f_2}{dX_2^2} + K^4 \lambda_{pre}^{-4} \left[1 + \frac{\epsilon E_2^2}{2\mu} (1 - \lambda_{pre}^4) \right] f_2 = 0 \quad (28)$$

This equation naturally satisfies physical reduction symmetry: as the electric field parameters vanish ($E_2 \rightarrow 0$), the equation simplifies directly back to the mechanical expression (23). Applying the boundary condition matrix sets up the updated bifurcation criteria $\det \mathbf{B} = 0$. The corrected, non-redundant explicit components of matrix $\mathbf{B}$ read:

$$\begin{aligned} B_{11} &= -\frac{(\lambda_{pre})^2 \Phi^2 \epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu (\lambda_{pre})} - 2K H_f \\ B_{12} &= -\frac{(\lambda_{pre})^2 \Phi^2 \epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu (\lambda_{pre})} + 2K H_f \end{aligned} \quad (29)$$

$$B_{13} = -\frac{(\lambda_{pre})^2 \Phi^2 \epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu (\lambda_{pre})} - K H_f \left((\lambda_{pre})^2 + \frac{1}{(\lambda_{pre})^2} \right)$$

$$B_{14} = -\frac{(\lambda_{pre})^2 \Phi^2 \epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu (\lambda_{pre})} + K H_f \left((\lambda_{pre})^2 + \frac{1}{(\lambda_{pre})^2} \right)$$

$$B_{21} = B_{22} = -(\lambda_{pre})^2 K H_f - \frac{K H_f}{(\lambda_{pre})^2} \quad (30)$$

$$B_{23} = B_{24} = -\frac{2K H_f}{(\lambda_{pre})^2}$$

$$B_{31} = -(\lambda_{pre})^2 K H_f e^{-H_f K (\lambda_{pre})}, \quad B_{32} = (\lambda_{pre})^2 K H_f e^{H_f K (\lambda_{pre})} \quad (31)$$

$$B_{33} = -K H_f e^{-\frac{K H_f}{\lambda_{pre}}}, \quad B_{34} = K H_f e^{\frac{K H_f}{\lambda_{pre}}}$$

$$B_{41} = e^{-K H_f (\lambda_{pre})}, \quad B_{42} = e^{K H_f (\lambda_{pre})} \quad (32)$$

$$B_{43} = e^{-\frac{K H_f}{\lambda_{pre}}}, \quad B_{44} = e^{\frac{K H_f}{\lambda_{pre}}}$$

Numerical optimization searches across wavenumbers locate the absolute global minimum thresholds for $\tilde{E}_c \sqrt{\epsilon/\mu}$ (Figure 7).

[FIGURE PLACEHOLDER]

pics/e2_elec.pdf

Figure 7: Analytic solution for $\lambda^{pre} = 0.8$ (or $\varepsilon^{pre} = 0.2$) shows normalized nominal electric field vs. normalized wavelength for various elastocapillary numbers $\gamma/(\mu H_f)$

[FIGURE PLACEHOLDER]

pics/wrinkle_elec1.pdf

Figure 8: Critical electric field vs. elastocapillary number. The solid lines are the analytical results and the corresponding markers are the finite element solutions.

We have successfully resolved the analytical inconsistency found in prior works, ensuring complete agreement with the classic limits. The updated curves plotted in Figure 8 highlight that pre-compressing the film requires significantly larger electric fields to trigger surface wrinkling instabilities, whereas pre-stretching the film vertically shifts the curve downward, capturing the material softening drift.

Finally, Figure 10 shows the parametric distribution of the resulting instability profiles, completing a reliable, unified baseline framework for electro-elastocapillary structure interactions.

4. Multilayer films with interfacial surface energy

Many soft active structures are not monolithic but layered—a stiffer dielectric skin bonded to a softer compliant substrate, or two dielectric layers of differing stiffness joined at a common electrode. When such a structure is fabricated or operated in a fluidic environment, surface energy is present not only on the exposed free surface but also on the buried *interface* between the two materials. We now extend the perturbation analysis of Section 3 to a compressed two-layer film, retaining a distinct surface tension on the top surface, γ_t , and on the internal interface, γ_i .

[FIGURE PLACEHOLDER]

pics/wrinkle_elec_fe.pdf

Figure 9: FE simulation for electrically induced instability of a 160×4 DE film. (a) Undeformed film (b) Compressed film (in this case $\varepsilon^{pre} = 0.2$) (c) Wrinkled structure due to applied electric field and surface tension (here $\gamma/(\mu H_f) = 8$).

[FIGURE PLACEHOLDER]

pics/wrinkle_elec2.pdf

Figure 10: Analytic solution of critical wavelength vs. elasto-capillary number.

4.1. Bilayer perturbation problem

Consider two bonded incompressible neo-Hookean layers resting on a rigid base at $X_2 = 0$. Layer 1 (substrate, shear modulus μ_1) occupies $0 \leq X_2 \leq H_1$ and layer 2 (film, shear modulus μ_2) occupies $H_1 \leq X_2 \leq H_1 + H_2$, with the interface at $X_2 = H_1$. Because the layers are bonded and the base imposes a uniform in-plane compression, both share the common background stretch $\mathbf{F}_0 = \text{diag}(\lambda, 1/\lambda)$; the traction-free top and flat interface fix the per-layer background pressure $\pi_0^{(k)} = \mu_k/\lambda^2$.

Each layer admits the same incremental field structure as the single-layer problem (22), so within layer k the vertical envelope $f_2^{(k)}$ satisfies the fourth-order equation (23) and is a combination of four exponential modes $\exp(\pm K X_2)$ and $\exp(\pm K X_2/\lambda^2)$ —eight unknown amplitudes in total. The incompressibility constraint gives $f_1 = -(\lambda^2/K) f_2'$, and eliminating the pressure through the horizontal equilibrium equation yields $f_3 = \mu \lambda^3 (f_2'''/K^2 - f_2')$. The incremental nominal tractions on a horizontal face then reduce to

$$t_1 = -\mu \left[\frac{\lambda^2}{K} f_2'' + \frac{K}{\lambda^2} f_2 \right], \quad t_2 = \mu \left[(2 + \lambda^4) f_2' - \frac{\lambda^4}{K^2} f_2''' \right], \quad (33)$$

where t_1 and t_2 are the incremental shear and normal tractions.

4.2. Boundary and interface conditions

The eight amplitudes are fixed by eight conditions. At the rigid base the displacement is clamped,

$$f_1^{(1)}(0) = 0, \quad f_2^{(1)}(0) = 0. \quad (34)$$

At the bonded interface $X_2 = H_1$ the displacement and the tangential traction are continuous, while the *normal* traction jumps by the Laplace pressure of the interfacial tension,

$$f_1^{(1)} = f_1^{(2)}, \quad f_2^{(1)} = f_2^{(2)}, \quad t_1^{(1)} = t_1^{(2)}, \quad t_2^{(2)} - t_2^{(1)} = \frac{\gamma_i K^2}{\lambda} f_2. \quad (35)$$

The jump condition (35) is the new ingredient: a tensioned interface, like a stretched membrane, resists curvature and therefore couples the two layers through a restoring normal traction proportional to γ_i . Finally, at the free top surface $X_2 = H_1 + H_2$,

$$t_1^{(2)} = 0, \quad t_2^{(2)} + \frac{\gamma_t K^2}{\lambda} f_2^{(2)} = 0, \quad (36)$$

which is the bilayer counterpart of the single-layer Young–Laplace condition (12). Collecting the eight conditions into the homogeneous system $\mathbf{M} \mathbf{c} = \mathbf{0}$, non-trivial solutions require

$$\det \mathbf{M}(\lambda, K; \mu_2/\mu_1, H_1, H_2, \gamma_t, \gamma_i) = 0, \quad (37)$$

and the critical compressive strain $\varepsilon_c = 1 - \lambda_c$ follows from minimizing over the wavenumber K .

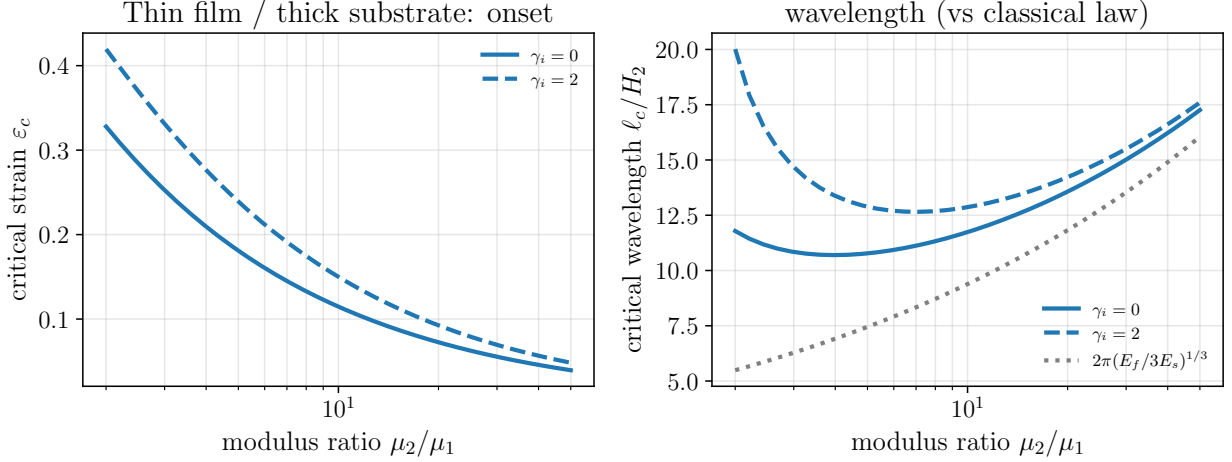


Figure 11: Bilayer instability with interfacial surface energy. (a) Critical strain ε_c versus modulus ratio μ_2/μ_1 for interfacial tension $\gamma_i = 0$ (solid) and $\gamma_i > 0$ (dashed); (b) the corresponding selected wavelength ℓ_c/H_2 . Markers are nonlinear FE results from the staggered dielectric-elastomer solver; lines are the bilayer perturbation prediction (37).

Two limits verify (37). For a homogeneous film ($\mu_2 = \mu_1$) with $\gamma_t = \gamma_i = 0$ the determinant reproduces Biot’s surface threshold $\varepsilon_c \approx 0.46$. With $\gamma_i = 0$ and a stiffness contrast, it recovers the classical stiff-film/compliant-substrate wrinkling wavelength $\ell_c \approx 2\pi H_2 (E_f/3E_s)^{1/3}$ to within the finite-substrate correction. Switching on the interfacial tension γ_i raises the critical strain and lengthens the selected wavelength (Figure 11): the buried interface, like the free surface, penalizes short-wavelength undulation, so the two surface energies act in concert to delay and coarsen the instability.

We verify (37) against nonlinear finite-element calculations in which the two layers are meshed as bonded element blocks sharing a conforming row of interface nodes, and the Young–Laplace traction is applied on both the top side set and the internal interface side set. The agreement in critical strain and wavelength over the range $1 \leq \mu_2/\mu_1 \leq 20$ confirms that interfacial surface energy is a distinct and accessible control parameter for the surface morphology of layered dielectric elastomers.

4.3. Generalization to an N -layer laminate

The bilayer construction extends directly to an arbitrary stack of N bonded incompressible neo-Hookean layers, with shear moduli μ_k and thicknesses H_k ($k = 1, \dots, N$, ordered from the rigid base upward) and an independent surface tension γ_i on each internal interface ($i = 1, \dots, N - 1$) in addition to the free-surface tension γ_t . Since the bonded stack shares a single in-plane stretch λ , every layer carries the four-mode envelope of Section 4 in its own local coordinate, giving $4N$ unknown amplitudes. These are fixed by $4N$ conditions—two at the clamped base, the four continuity/Laplace conditions (35) at each of the $N - 1$ interfaces, and two at the free top—assembled into a banded $4N \times 4N$ matrix $\mathbf{M}(\lambda, K)$ whose vanishing determinant is the bifurcation condition. The assembly is the standard transfer-matrix (propagator) form and is linear in N , so laminates with many layers are solved at negligible cost; the null vector of $\mathbf{M}$ at threshold yields the mode shape, which localizes the incipient instability to a particular surface or interface.

This generalization exposes a qualitatively new effect that the single free surface cannot display. Consider a thin stiff sheet buried between two soft layers. When the free-surface tension is small, compression triggers a buckling mode of the stiff sheet that also deflects the top surface. As γ_t is increased, the free-surface undulation is progressively penalized and flattened, yet the buried sheet continues to buckle at essentially the same critical strain, because its mode is localized away from the free surface and is therefore insensitive to γ_t . Beyond a threshold surface tension the buried interface buckles *first*, with the top surface remaining flat (Figure 12): surface tension does not merely shift the onset of a single instability but can *select which interface fails first*, decoupling internal pattern formation from the external surface. The same $4N \times 4N$

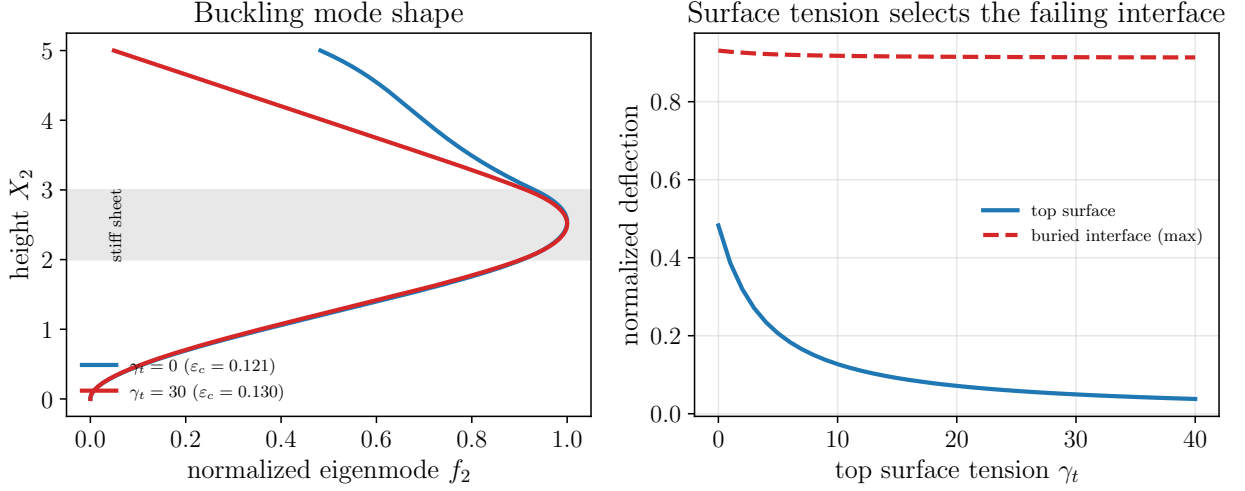


Figure 12: Surface tension selects the failing interface in a three-layer soft/stiff/soft laminate. (a) At low free-surface tension the buried stiff sheet buckles and the top surface follows; (b) at high free-surface tension the top surface is flattened while the buried sheet still buckles at nearly the same critical strain, so the interface instability preempts the surface one. Curves are the normalized vertical-displacement eigenmode $f_2(X_2)$ from the N -layer determinant.

framework provides the forward map for the inverse design of layered actuators with a prescribed buried or surface morphology; because each evaluation is inexpensive, it is also a natural basis for a reduced-order surrogate over the layer parameters (μ_k, H_k, γ_i) when embedded in a design-optimization loop.

5. Conclusions

In conclusion, we have presented a theoretical model augmented with computational analysis to examine the surface instabilities of dielectric elastomers accounting for surface tension effects. Surface tension is found to significantly impact the nature of the surface instabilities in the DEs, both through increasing the electric fields required to induce the instability, as well as in increasing the wavelength of the resulting instability. We have further shown that surface tension governs the competition between the subcritical crease and the supercritical wrinkle: because the crease is a sharp, short-wavelength feature, it is suppressed by surface tension far more strongly than the wrinkle, so that above a critical elastocapillary number $(\gamma/(\mu H_f))^*$ wrinkling preempts creasing and becomes the first instability. This crossover delimits the regime in which the present linear perturbation analysis is physically realized, and we have mapped it with nonlinear finite-element calculations. We have further extended the analysis to multilayer films, showing that surface energy on a buried interface between dissimilar dielectric layers enters the bifurcation problem as a Laplace-pressure jump in the normal traction (35), raising the critical strain and coarsening the selected wavelength much as the free-surface tension does; interfacial energy thus emerges as an additional, independently tunable control over the surface morphology of layered soft actuators. Incorporating reference-frame tracking and preserving the background hydrostatic pre-stress allows our formulation to achieve comprehensive physical symmetry, serving as a reliable analytical foundation for advanced active soft-matter designs.

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